

Homework #2

100 Points Possible

7/1/2026

Attempt 1



In Progress

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Technical Indicators & Strategy Backtesting with Alpaca

Objective

Using historical market data from Alpaca, build a backtesting platform and evaluate multiple algorithmic trading strategies.

The goal is to answer:

Which strategy performs best on a risk-adjusted basis?

Requirements

1. Historical Data

Using Alpaca's Historical Market Data API:

- Download at least 5 years of daily OHLCV data
- Allow the user to select a ticker
- Store the data in a Pandas DataFrame

Examples:

AAPL
MSFT
SPY
QQQ
NVDA

2. Technical Indicators

Implement at least **6 indicators** from the list below:

Trend

- SMA
- EMA
- MACD
- ADX

Momentum

- RSI
- Stochastic Oscillator
- Williams %R

Volatility

- Bollinger Bands
- ATR

Volume

- OBV
 - Chaikin Money Flow (CMF)
-

3. Trading Strategies

Create and compare the following strategies:

Strategy 1: Trend Following

Use indicators such as:

- MACD
- ADX
- Moving Averages

Example:

Buy when:

- $MACD > Signal$
- $ADX > 25$

Sell when:

- $MACD < Signal$
-

Strategy 2: Mean Reversion

Use indicators such as:

- RSI
- Bollinger Bands

Example:

Buy when:

- $RSI < 30$
- Price below lower Bollinger Band

Sell when:

- $RSI > 70$
 - Price above upper Bollinger Band
-

Strategy 3: Custom Strategy

Design your own strategy using **at least three indicators**.

Your strategy must combine indicators from at least **two different categories** (trend, momentum, volatility, volume).

4. Backtesting Engine

Build a reusable backtesting engine.

Assumptions:

- Initial Capital: \$100,000
- Long-only
- No leverage
- No short selling

Track:

- Portfolio value
 - Daily returns
 - Trades executed
-

5. Performance Metrics

Calculate:

- Total Return
 - CAGR
 - Volatility
 - Sharpe Ratio
 - Sortino Ratio
 - Maximum Drawdown
 - Win Rate
-

6. Visualizations

Create the following charts:

Price Chart

Show:

- Price
- Indicators used
- Buy/Sell signals

Equity Curve

Compare:

- Buy & Hold
- Strategy 1
- Strategy 2
- Strategy 3

Drawdown Chart

Compare drawdowns for all strategies.

7. Final Report

Include:

- Strategy descriptions
- Entry and exit rules
- Performance comparison table
- Discussion of results

Example:

Strategy	Return	Sharpe	Sortino	Max Drawdown
Buy & Hold				
Trend Following				
Mean Reversion				
Custom Strategy				

8. GitHub Submission

Your repository must include:

- README.md
- Source code
- requirements.txt
- Charts
- Final report (PDF)

Grading Rubric (100 Points)

Component	Points
Alpaca data retrieval	10
Technical indicators	15
Strategy 1 (Trend Following)	15
Strategy 2 (Mean Reversion)	15
Strategy 3 (Custom)	15
Backtesting engine	10
Performance metrics	5
Visualizations	5
Final report	3
Video	5
GitHub repository	2

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